Cozy Bear Cottage 小熊中文

Meta Global AI Developer Hackathon · Indie Track · An AI co-teacher for weekend Chinese classes, ages 3–6

The problem in one line

A weekend Chinese teacher spends her Friday evening building one 45-minute lesson by hand — words, story, songs, games, a script — for children who learn by touching, hearing, and repeating.

Cozy Bear builds that lesson in one step, then lives inside it with the children.

Hackathon build: designed, built, and launched live during hackathon week, October 26–31, 2026 — start to finish.

Live app: **cozybear-cottage.vercel.app**

Code: github.com/hellohelloye/cozy-bear-cottage-app

OUR TEAM

Team Lead: **Yukui McDonald**

Team members:

Patrick McDonald

Xiaoyue He

Shun Zhang

How this project answers the rubric

Official judging criteria — and where each one is proven

IMPACT — 30%	**Slides 3**	A real Saturday class, a real teacher, real children — not a hypothetical user
META AI TOOLING — 25%	**Slide 4**	Muse Spark, Muse Image, and SAM 3.1 carry the core features
DEMO — 25%	**Slide 5**	A 3-minute guided path through the live app; every beat has a fallback
TECHNICAL DEPTH — 20%	**Slides 6–9**	Serverless proxy, strict-JSON contracts, graceful degradation, secrets discipline — plus both architecture diagrams

Judging is submission-based: project write-up, live app link, team roster, and demo video. Submission due Nov 4, 2026, 9:00 PM ET.

IMPACT — 30%

A working class, not a demo persona

- **It already runs:** a Saturday-morning Chinese class for ages 3–6 meets weekly in Bear, Delaware. Cozy Bear was built for that room, by its teacher.
- **Teacher time returned:** one theme in → a complete 45-minute lesson out (words, story, games, teaching script, songs) — replacing an evening of manual prep, every week.
- **Built for how 3–6-year-olds actually learn:** tap a picture, hear the word spoken in Mandarin, say it back, earn a star. Big targets, instant sound, no reading required.
- **The children's own art is the curriculum world:** the pictures children explore in Point-and-Say are paintings by the class's own young artists — Vivian (4) and Evelyn (3) — so students learn inside art made by kids like them.
- **Families stay connected:** the Kids' Artwork gallery lets children add their own artwork to the cottage, and artwork can be emailed to the teacher.

USE OF META AI TOOLING — 25%

Three Meta models, three jobs

All reached through one serverless proxy — the API key never touches the browser

Muse Spark 1.3	Lesson Builder	Turns a theme + age range into a complete lesson as strict, validated JSON — words, script, songs, activities
Muse Spark 1.3	Point-and-Say	Vision naming: given a photo and a tap point, returns the object's Chinese name, pinyin, English, and bounding box
Muse Image 1.0	Living Flashcards	Draws a fresh, child-friendly illustration for every word of the week
SAM 3.1	Point-and-Say	Segments the tapped object and tracks it frame-by-frame in short clips (flow integrated; a curated pack of 26 pictures / 57 tap zones serves today's live taps)
Web Speech API	Everywhere	zh-CN text-to-speech speaks each word aloud at child pace (rate 0.85); spoken karaoke in the Sing-along

Path: browser → POST /api/meta (Vercel function; META_API_KEY is a server secret) → Meta Model API at api.meta.ai/v1.

DEMO — 25%

The 3-minute demo path

Recorded on the live app — every beat below is demonstrable today

0:00	Meet the team and the class — a real Saturday class in Bear, Delaware
0:25	Lesson Builder: pick Colors → Muse Spark builds the full lesson live
1:10	Sing-along + Living Flashcards: the theme song plays; Muse Image draws each word's card
1:45	Point-and-Say: tap the dog — 狗! — spoken aloud, star awarded
2:20	Kids' Artwork: the gallery, the young artists at work, a classmate's upload
2:45	About: the story, the team, the class poster — and the close

Designed to never dead-end: sample lesson, emoji cards, and the curated photo pack stand behind every live model call.

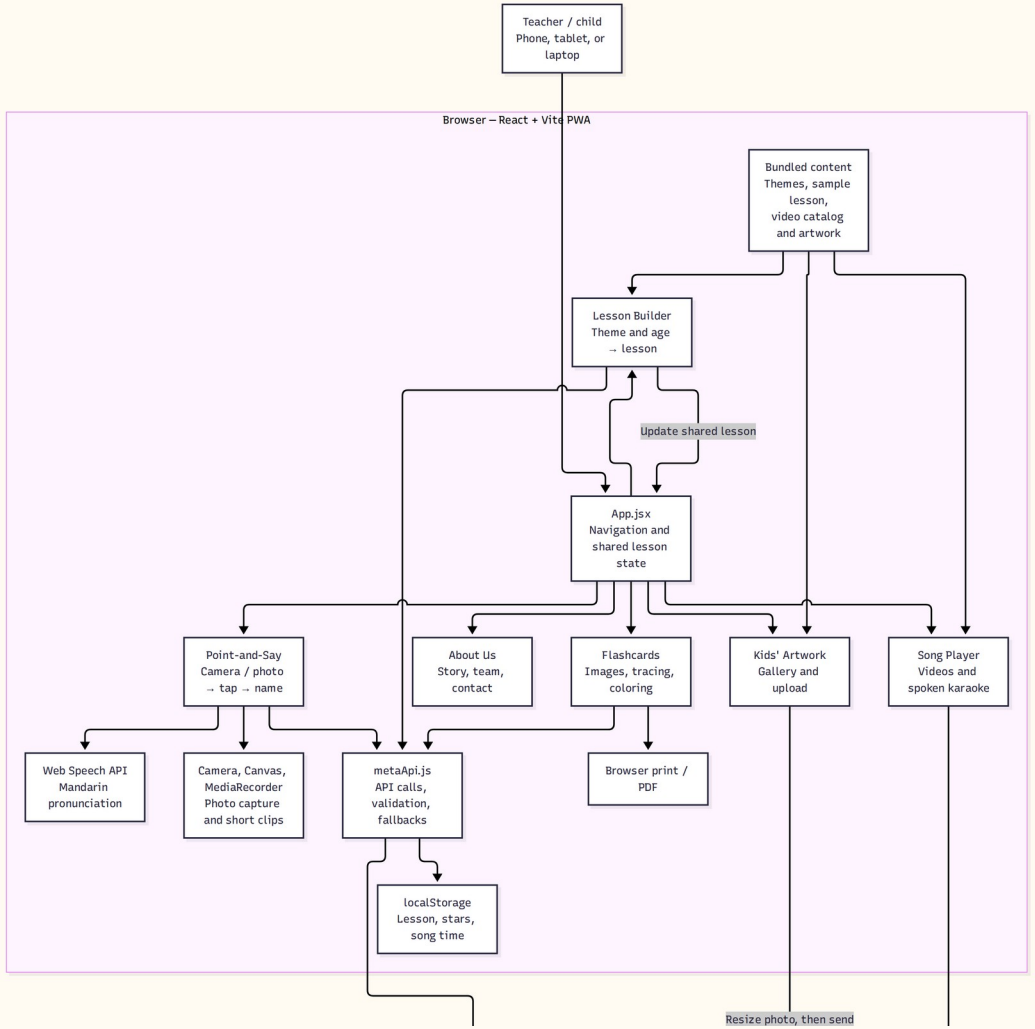
TECHNICAL DEPTH — 20%

Small surface, real engineering

- **Architecture:** React 18 + Vite PWA on Vercel; App-level shared lesson state carries one generated lesson through flashcards, songs, and Point-and-Say.
- **Contract-first AI:** the proxy validates the request, dispatches to the right model, and parses responses into strict shapes before the UI sees them.
- **Point-and-Say pipeline:** tap → segmentation (SAM 3.1) → naming (Muse Spark, with a label-based fallback chain) → Mandarin TTS → star; short clips add per-frame tracking to refine the box.
- **Graceful degradation:** every AI feature has a local fallback — bundled sample lesson, emoji flashcards, pre-mapped tap zones — so a stalled model call never breaks the class.
- **Secrets discipline:** META_API_KEY and RESEND_API_KEY exist only as Vercel server secrets; artwork email runs through a second function, POST /api/artwork, via Resend.
- **State that survives the classroom:** lessons, stars, and song position persist in localStorage; the app installs as a PWA on the class tablet.

TECHNICAL DEPTH — 20%

System architecture (1 of 2) — inside the browser



What to notice

Everything the child touches lives in the browser: the five tabs, bundled content, camera, and Mandarin speech.

App.jsx holds the shared lesson state — one generated lesson flows into flashcards, songs, and Point-and-Say.

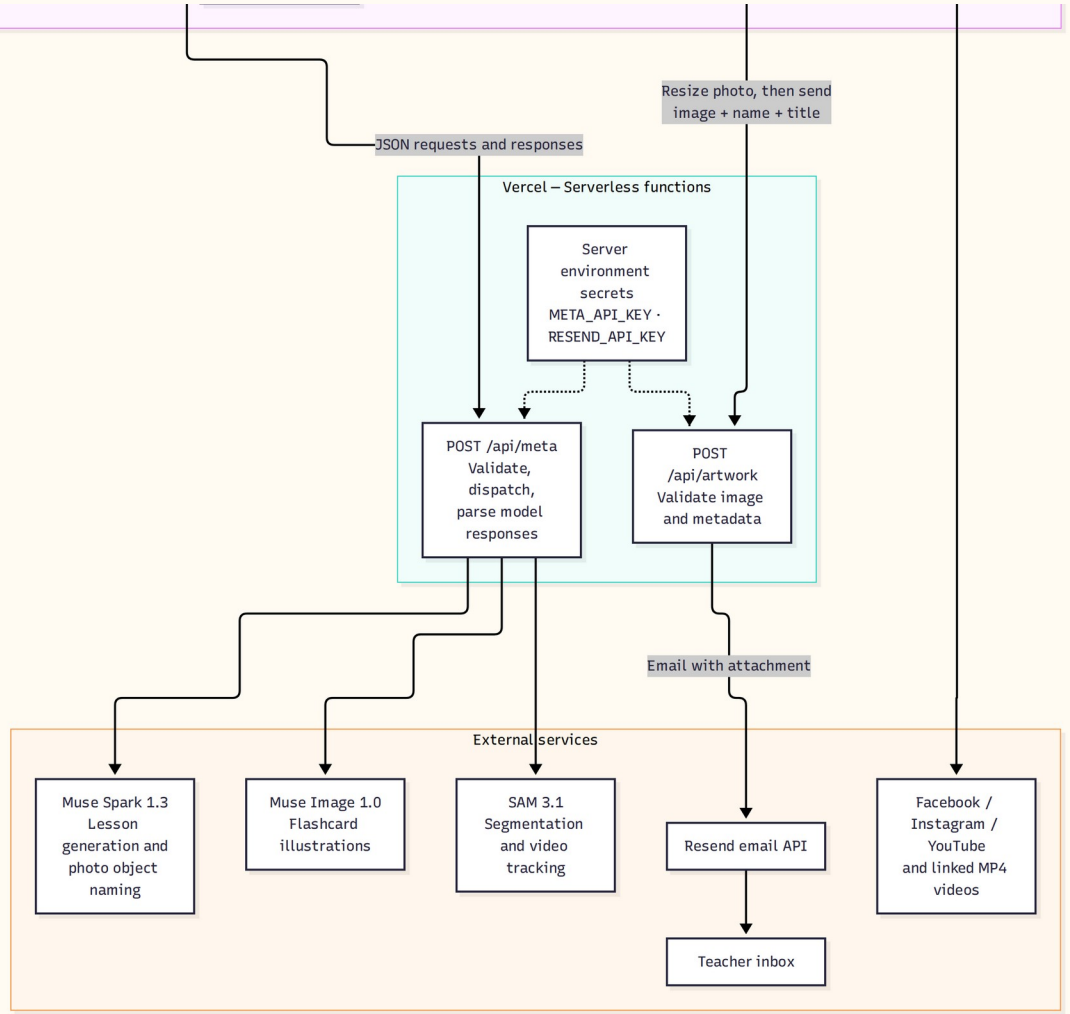
metaApi.js is the single door to the models: every AI call, validation, and fallback passes through it.

localStorage keeps the lesson, stars, and song position between Saturdays.

Diagrams also live in the GitHub README as native Mermaid — judges can zoom; every label stays sharp.

TECHNICAL DEPTH — 20%

System architecture (2 of 2) — proxy & Meta models



What to notice

Two functions, two secrets: **META_API_KEY** and **RESEND_API_KEY** exist only on the server.

POST /api/meta validates each request, dispatches to the right model, and parses the response before the UI sees it.

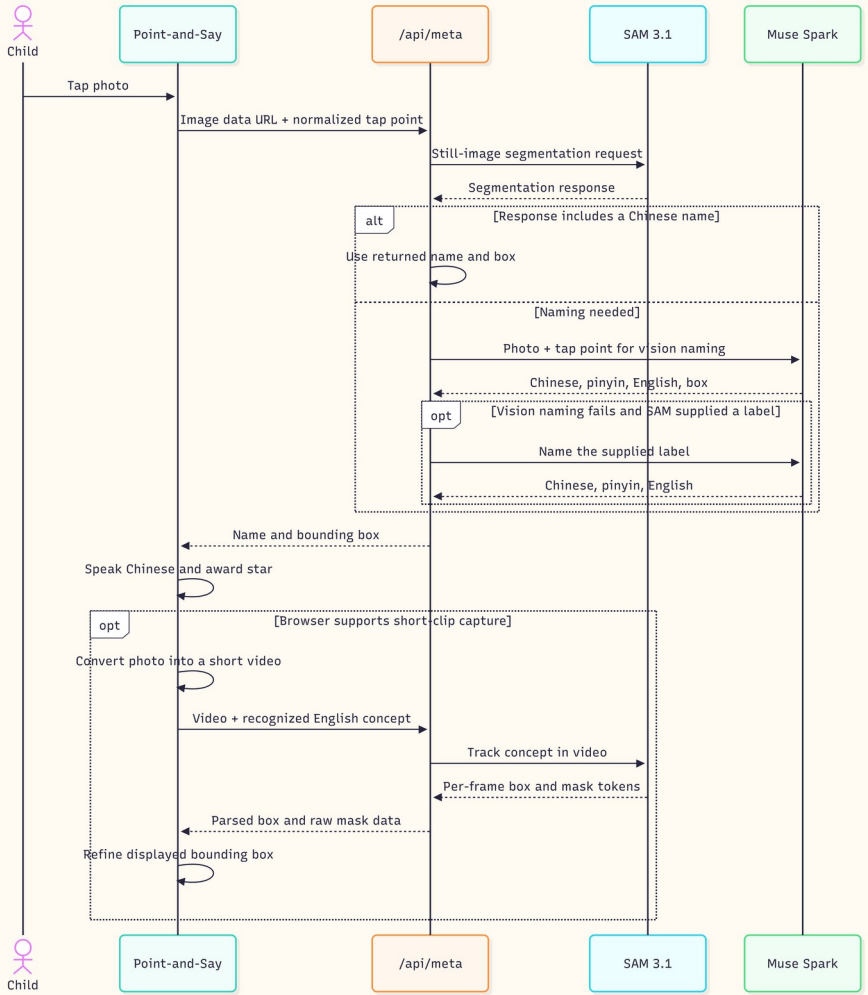
POST /api/artwork emails children's artwork to the teacher through Resend.

Below the line: **Muse Spark 1.3, Muse Image 1.0, SAM 3.1** — and the video platforms behind the Sing-along.

Diagrams also live in the GitHub README as native Mermaid — judges can zoom; every label stays sharp.

TECHNICAL DEPTH — 20%

Point-and-Say — the tap-to-name sequence



What to notice

One tap crosses four systems: **SAM 3.1** segments the tapped object first.

If no Chinese name comes back with the segment, **Muse Spark** names it from the photo — and a second fallback names SAM's own label if vision naming fails.

The name returns with its box; the app speaks it in Mandarin and awards a star.

Short clips add **SAM 3.1 video tracking**, refining the box frame by frame.

Diagrams also live in the GitHub README as native Mermaid — judges can zoom; every label stays sharp.

The five tabs

One lesson, five ways to live in it

1 · Lesson Builder 备课

Theme + age in → full lesson out: words of the week, story, games, a word-for-word teaching script, and the Sing-along player with spoken karaoke over the Chinese Playtime catalog.

2 · Living Flashcards 单词卡

Each word becomes a card with a Muse Image illustration; tracing and coloring modes; a print pack for the classroom wall.

3 · Point-and-Say 指着说

Tap anything in a picture — it is segmented, named in Chinese / pinyin / English, and spoken aloud. Repeat it, earn a star. Short clips refine the highlight frame by frame.

4 · Kids' Artwork 孩子们的画

Vivian's and Evelyn's paintings are the world children explore; classmates upload their own artwork with a title and artist name, and can email it to the teacher.

5 · About 关于我们

The story, the team, the Saturday class poster, and the cottage inbox: applebearscabinet26@gmail.com.

AI that teaches like a teacher

Cozy Bear Cottage — built in a real classroom, for real children. Thank you.

OUR TEAM

Team Lead: **Yukui McDonald**

Team members:

Patrick McDonald

Xiaoyue He

Shun Zhang

Artwork by our kids:

Vivian McDonald · visual design & artwork

Evelyn McDonald · artwork

SUBMIT BY

Nov 4, 2026 · 9:00 PM ET

Live app

cozybear-cottage.vercel.app

Code

github.com/hellohelloye/cozy-bear-cottage-app

Contact

applebearscabinet26@gmail.com